

Elle Buser

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EDUCATION

Emory University, Atlanta, GA

August 2022 – Present

Ph.D. in Computational Mathematics

Masters in Computational Mathematics

University of Wyoming, Laramie, Wyoming

June 2018 – May 2022

Bachelor of Science in Mathematics, Minor in Physics

Research Interests

I am interested in computational inverse problems, uncertainty quantification, and numerical linear algebra. I am currently working on efficient methods to overcome challenges in parameter estimation for large-scale Bayesian inverse problems by using stochastic estimators, minimization-majorization algorithms, and natural gradient descent.

PUBLICATIONS

- [1] Elle Buser, Julianne Chung, Hugo Diaz, and Arvind Saibaba
“A Majorization-Minimization with Monte Carlo Approach for Hyperparameter Estimation.” Submitted, 2026.
<https://arxiv.org/abs/2605.13620>
- [2] Eric Zou, Elle Buser, Zichao (Wendy) Di, and Yuanzhe Xi
“A Joint Variational Framework Framework for Multimodal X-ray Ptychography and Fluorescence Reconstruction.” Submitted, 2025. [arXiv:2511.02153v1](https://arxiv.org/abs/2511.02153v1)
- [3] Elle Buser and Julianne Chung
“Efficient Sampling Approaches Based on Generalized Golub-Kahan Methods for Large-Scale Hierarchical Bayesian Inverse Problems.” Submitted, 2025. [arXiv:2502.03281](https://arxiv.org/abs/2502.03281)
- [4] Elle Buser, Emma Hart, Ben Huenemann
“Comparison of atlas-based and neural-network-based semantic segmentation for DENSE MRI images.” SIAM Undergraduate Research Online (SIURO), 15, 2022. [PDF](#)
- [5] Will T. Chick et al.
“The Wind beneath My Wings. I. Spectral Types and Multiplicity of the Central Stars Supporting Stellar Bow Shock Nebulae.” The Astrophysical Journal Supplement Series, 251(2), 2020.

RESEARCH

Emory University

Fall 2022 – Present

Advisor: Dr. Julianne Chung

Developing robust and efficient computational methods for reconstructions in medical imaging, atmospheric modeling, and seismic inversion with focuses on parameter estimation and uncertainty quantification

Argonne National Lab

Summer 2024

Advisor: Dr. Zichao (Wendy) Di

Worked to recover high-resolution structural and compositional information from X-ray Ptychography and X-ray Fluorescence measurements by using a joint variational framework

NSF REU at Emory University

Summer 2021

Advisor: Dr. Lars Ruthotto

Implemented a neural-network-based image segmentation model to automate the process of identifying the brainstem and cerebellum in the DENSE MRIs of potential Chiari patients

University of Wyoming

Spring 2019 – Spring 2020

Advisor: Dr. Henry (Chip) Kobulnicky

Wrote Python codes which produced and predicted light curves of a binary system from online data bases and modeled flux and radial velocity of candidate bow shock nebula supporting binary stars

POSTERS AND PRESENTATIONS

(Invited) Spring Research Conference on Statistics in Engineering, Science and Technology, Clemson, SC. “Natural Gradient Descent for Hyperparameter Estimation in Bayesian Inverse Problems” May 2026.

(Invited) 27th Conference of the International Linear Algebra Society, Blacksburg, VA. “Natural Gradient Descent for Hyperparameter Estimation in Bayesian Inverse Problems” May 2026.

(Invited) SIAM Conference on Uncertainty Quantification, Minneapolis, MN. “Computational Methods For Hyperparameter Estimation with Applications to Separable Nonlinear Inverse Problems” March 2026.

ICERM Workshop on Bayesian Inverse Problems and UQ, Providence, RI. “Computational Methods For Hyperparameter Estimation with Applications to Separable Nonlinear Inverse Problems” March 2026.

Discussions in Scientific Computing Graduate Seminar, Emory University. “Computational Methods For Hyperparameter Estimation with Applications to Separable Nonlinear Inverse Problems.” November 2025

(Invited) SIAM Annual Meeting, Montreal, Canada. “Generalized Golub-Kahan Methods for Large-Scale Hierarchical Bayesian Inverse Problems.” July 2025.

Georgia Scientific Computing Symposium, Georgia Institute of Technology, Atlanta, GA. “Generalized Golub-Kahan Methods for Large-Scale Hierarchical Bayesian Inverse Problems.” February 2025.

Argonne National Lab, Virtual. “Joint Analysis of X-ray Ptychography and X-ray Fluorescence.” August 2024

(Invited) SIAM Conference on Imaging Science, Atlanta, GA. “Krylov Subspace Methods for Large-Scale Hierarchical Bayesian Inverse Problems.” May 2024.

Discussions in Scientific Computing Graduate Seminar, Emory University. “Efficient Krylov subspace methods for large-scale hierarchical Bayesian inverse problems.” April 2024

Discussions in Scientific Computing Graduate Seminar, Emory University. “Projection Based Reduced Order Modeling for Parametric Systems.” November 2022

Georgia Scientific Computing Symposium, Georgia Institute of Technology, Atlanta, GA. “Comparison of atlas-based and neural-network-based semantic segmentation for DENSE MRI images.” February 2022.

TEACHING

Emory University

MATH 112: Calculus II, Instructor of Record	Fall 2025 – Spring 2026
MATH 111: Calculus I, Instructor of Record	Fall 2024 – Spring 2025
MATH 221: Linear Algebra, TA	Fall 2023 – Spring 2024
MATH 351: Partial Differential Equations, Grader	Fall 2022 – Spring 2023

University of Wyoming

MATH 2200: Calculus I, Tutor	Spring 2021 – Spring 2022
MATH 2205: Calculus II, Tutor, Supplemental Instructor	Spring 2021 – Spring 2022
PHYS 1210: Engineering Physics I, Tutor	Spring 2021 – Spring 2022

AWARDS

Student Travel Award, ASA/IMS Spring Research Conference	2026
SIAM Student Travel Award, NSF	2026
Women in Natural Science Fellowship, Emory University	Fall 2022 – Present
Wyoming Research Scholar, University of Wyoming	Spring 2019 – Spring 2020
Rocky Mountain Scholar, University of Wyoming	Fall 2018 – Spring 2022

OUTREACH

SIAM Student Chapter Treasurer, Emory University, Atlanta, GA	Fall 2025 – Spring 2026
Co-Organizer SIAM AN25 Mini Symposium, Montreal, Canada	July, 2025
Emory Directed Reading Program Mentor, Emory University, Atlanta, GA	Spring 2025
Atlanta Science Festival Volunteer, Atlanta, GA	Spring 2025
STEAM Night at Mary Lin Elementary School Volunteer, Atlanta, GA	Fall 2025
Math Circle Middle School Instructor, Emory University, Atlanta, GA	Fall 2023, Fall 2024
Julia Robinson Mathematics Festival Volunteer, Emory University, Atlanta, GA	Fall 2022, 2023, 2024

TECHNICAL SKILLS

MATLAB, Python, L^AT_EX

ADDITIONAL CONFERENCES AND WORKSHOPS

Gene Golub SIAM Summer School, Quito, Ecuador. Summer 2024.

Copper Mountain 18th Conference on Iterative Methods, Copper Mountain, CO. April, 2024.

AMS Southeastern Sectional Meeting, Georgia Institute of Technology, Atlanta, GA. March 2023.

Georgia Scientific Computing Symposium, Georgia State University, Atlanta, GA. February, 2023.

SIAM Conference on Mathematics of Data Science, San Francisco, CA (virtually). September, 2022.